Talk 1

Tuesday, March 10, 2026 4:00 PM

what we want to do is to study group representations $\mathrm{Rep}(G)$. So, what is $\mathrm{Rep}(G)$

Def: the category of linear representation of a group G is the category with the following,

1) obj: V -, with G-action $\rho: G \longrightarrow GL(V)$
2) morphisms: linear maps, G-interwiners

$$f(g \triangleright v) = g \triangleright f(v).$$

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if we look at the space we see that it has a k-algebra structure we can add stuff, multiply and so on.

but G only allows for multiplication.

To resolve the mismatch we force addition on G over our field k. basically, we form the group algebra

$$k[G] \qquad k[G] \ni x = \sum_{g \in G} c_g g$$

multiplication in $k[G]$ is induced by multiplication in G.

But how would this help us study $\mathrm{Rep}(G)$.

So, how would this help us study Group rep?

a rep $\rho: G \longrightarrow GL(V) = \mathrm{End}(V)$

($GL(V)$ is a group)

equivlent

$$\Rightarrow \quad \tilde{\rho}: k[G] \longrightarrow \mathrm{End}(V)$$

we define a $k[G]$ module str on V

$$k[G] \times V \longrightarrow V$$

just by

$$x \cdot v = \sum c_g \rho(g) v$$

(clearly satisfies associativity axiom)!

(the inverse is clear)

$$\rho(g) \cdot (v) = g \triangleright v \quad \text{module action!}$$

And then this directly implies that an interwiver induces a module morphism so, we get

$$\mathrm{Hom}_{k\text{-Alg}}(k[G], A) \simeq \mathrm{Hom}_{\mathrm{Grp}}(G, U(A))$$

$$\mathrm{Grp} \xrightarrow{k[\,]} k\text{-Alg}$$

$\rho_V \leftarrow \rho_U$

This just means:

$$\mathrm{Rep}(G) \simeq \underset{K[G]}{\mathcal{M}}$$

We want to translate everything we know about $\mathrm{Rep}(G)$ to $\underset{K[G]}{\mathcal{M}}$

- Monoidal structure: if $V, W \in \mathrm{Rep}(G)$
  $\Rightarrow V \otimes_K W \in \mathrm{Rep}(G)$

$$\rho_{V\otimes W}(g)\,(v\otimes w) = \rho_V(g)\,v \otimes \rho_W(g)\,w$$

$$\Rightarrow \quad g\cdot(v\otimes_K w) = (g\cdot v)\otimes(g\cdot w)$$

The only thing we know here is that if $V, W \in \underset{K[G]}{\mathcal{M}}$ then $V\otimes W$ is a $K[G]\otimes K[G]$ module

So $\longrightarrow$ demand an algebra map

$$\Delta : K[G] \longrightarrow K[G]\otimes K[G]$$

s.t. $\Delta(g) = g\otimes g$

we also have a unit $V\otimes 1 = V$

Trivial rep

$x \in K$, $\rho(g)\cdot x = x$

so, just make a map $\varepsilon : K[G] \longrightarrow K$

s.t.

$$\varepsilon(g)\,x = x \quad \Rightarrow \quad \varepsilon(g) = 1$$

- Rigidity:

For every rep we have V^*, V finite dimensional

So, we have $f \in V^*$

$$(\rho_{V^*}(g))(f)(v) = f\left[\rho_V(g^{-1})\cdot v\right]$$

notice having the inverse is strictly necessary we would not have associative action otherwise

$$\rho(gh)(f)(v) = f\left(\rho(gh)^{-1})\cdot v\right) = f\left(\rho(h^{-1})\cdot\rho(g^{-1})\cdot v\right)$$
$$= \rho(g)\,\rho(h)\,(f)\,(v)$$

$\Rightarrow$ we want V^* to be a $K[G]$-module

So, for the same reason we have

$(\rho\otimes\rho)(x)$ $\rho(x^{-1})$ $\rho(x^{-1})$

$(g \cdot f)(x) = f(\underbrace{S(g)}_{g^{-1}} \cdot x) = f(g^{-1} \cdot x)$

s.t $S(g) = g^{-1}$

with S being an antihomeomorphism map from $k[G] \to k[G]$. $\quad S(xy) = S(y)\,S(x)$

Now, we have all the structure on $k[G]$. let us axiomatize this!

first one can easily check the following relations

- $(\varepsilon \otimes id)\,\Delta = (id \otimes \varepsilon) \circ \Delta = id$
- $\mu(S \otimes id) \circ \Delta(x) = \mu \circ (id \otimes S) \circ \Delta(x) = \varepsilon(x)$
- Δ is coassociative

$$(id \otimes \Delta) \circ \Delta = (\Delta \otimes id) \circ \Delta$$

Def: A quantum group or Hopf algebra is unital associative algebra A which is equipped with (Δ, ε, S) and has the above properties. (without antipode it's called a bialgebra).

Examples: . $k[G]$ is indeed a hopf algebra

⊙ Interesting example: let g be a Lie algebra

Consider $U(g)$ it's universal enveloping algebra

$\{$ words of elements in $g\}$ / $\{$commutations relations$\}$

we have

$$\Delta(x) = x \otimes 1 + 1 \otimes x$$
$$S(x) = -x$$
$$\varepsilon(x) = 0$$

} x is called primitive

$U(g)$ is cocommutative.

but we want something which is not cocommutative

⊙ Protopy quantum group $U_q(sl_2)$

$q \in \mathbb{C} \qquad q \neq 0, \pm 1$

$$U_q(sl_2) = \left\langle e, f, k^{\pm 1} \;\middle|\; kek^{-1} = q^2 e \quad kfk^{-1} = q^{-2} f \right\rangle$$
$$ef - fe = \frac{k - k^{-1}}{q - q^{-1}}$$

with, $\Delta(e) = e \otimes k + 1 \otimes e \qquad \varepsilon(e) = \varepsilon(f) = 0$

$\Delta(f) = f \otimes 1 + k^{-1} \otimes f \qquad \varepsilon(k) = 1$

$\Delta(k) = k \otimes k$

$S(e) = -ek^{-1} \qquad S(f) = -kf, \quad S(k) = k^{-1}$

(can be found from hopf algebra relation)

Rmk: you go back to $U(sl_2)$ by taking $q = t^{h/2}$

$k = q^h$ and sending $t \to 0$.

Question! in the perivous example like Rep(G) it was easy to guess the structure of the hopf alg $k[G]$. but suppose I give you a category $\mathcal{C}$ which we know $\simeq A$-mod

Can you find A?

(Notice this doesnot hand us A on a sliver plate)

We notice that the action of the algebra A must commute with an $f: M \to N$. (let us denote the action of A on module M by η_M)

So, we get

$$f \circ \eta^a_N = \eta^a_M \circ f$$

where $\eta^a_M : M \to M$ action defined by element $a \in A$

more precisely for each $a \in A$, we get a map $a_M \in End(M)$ as over for space s.t

$$\begin{array}{ccc} M & \xrightarrow{a_M} & M \\ f\downarrow & & \downarrow f \\ N & \xrightarrow[a_N]{} & N \end{array} \quad \text{commutes}$$

So, intuitively speaking, we just associated an element of the algebra to a map in $End(M)$. but these are the space of homeomorphisms of the vector space right. So this motivates the following

(Def: let F be a functor $F: \mathcal{C} \to Vec$ which forgets the action of A.

Then from above

$$\begin{array}{ccc} F(M) & \xrightarrow{a_M} & F(M) \\ \downarrow & & \downarrow \\ F(N) & \longrightarrow & F(N) \end{array}$$

(a_M defined above)

backward direction

Given $\eta \in End(F)$

$\Rightarrow \eta_A(1_A)$ is our claim

You can check that these maps are inverses

$$A \simeq End(F) \leftarrow \text{natural transformations of } F.$$

Rmk: this is just Yoneda lemma by noticing that

$$Hom_A(A, M) \simeq M \simeq F(M) \leftarrow F \text{ is representable}$$

$$\Rightarrow Nat(Hom_A(A, -), F) = F(A)$$

$$\Rightarrow End(F, F) \simeq F(A) \simeq A$$

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ok now what? How can we recover the original

Ok now we have, now can we recover the coproduct:

a coproduct has to be

$$\Delta : A \longrightarrow A \otimes A$$

$$\rightsquigarrow \quad End(F) \longrightarrow End(F) \otimes End(F) \simeq End(F \otimes F)$$

$$F \otimes F : \mathcal{C} \otimes \mathcal{C} \longrightarrow Vec$$
$$X \otimes Y \longrightarrow F(X) \otimes F(Y)$$

So, we must identify for element in $A \otimes A$ an operator in $End(F \otimes F)$ basically a map:

$$\Delta(a)_{X,Y} : F(X) \otimes F(Y) \longrightarrow F(X) \otimes F(Y)$$

to define this map we need to use information in a!
the only way to do so is to impose a monoidal str on the functor!

- Def: $F(X) \otimes F(Y) \xrightarrow{J_{X,Y}} F(X \otimes Y) \xrightarrow{F(a_{X \otimes Y})} F(X \otimes Y) \xrightarrow{J^{-1}_{X,Y}} F(X) \otimes F(Y)$

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one other technical construction we need also the reconstruct category Δ-mod to have same same submodule str. This translate into a relation on exact sequence in $\mathcal{C}$ if

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0 \text{ is exact} \Rightarrow 0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z) \longrightarrow 0 \text{ is exact!}$$

In that way we just made an equivalence

Prop: H is ass unital Alg and $\mathcal{C} \simeq H\text{-mod}$,
Suppose $\mathcal{C}$ is equipped with a monidal str, $F : \mathcal{C} \longrightarrow Vec$ be exact, monoidal functor (fiber functor)
faithful

$\Rightarrow H \simeq End$ is a bialgebra
is hopf if $\mathcal{C}$ is rigid!

F is an equivalence $\mathcal{C} \simeq Rep(H)$

Rmk: on rigidity, because of the evaluation map

$$X^* \otimes X \longrightarrow K$$

we get a pairing non degen

$$F(X^*) \otimes F(X) \longrightarrow K \quad \Rightarrow \quad F(X^*) \simeq F(X)^*$$

we know that for $a \in End\, F$ $\quad a_{X^*} : F(X) \longrightarrow F(X^*)$
but it has also to be a map $: F(X)^* \longrightarrow F(X)^*$

$$\Rightarrow \quad a_X : F(X) \longrightarrow F(X) \quad \Rightarrow \quad S(a)_X = (a_{X^*})^*$$
(transpose)

then you can check the relations!

Rmk: equivalence between fiber functors ⇒ twist equivalence of hopf algebras (coinfold twist).

Now let us look back again at these rigid monoidal categories

$\mathcal{C}$ which has objects X and some morphisms



, monoidality



rigidity, $ev : X^* \otimes X \longrightarrow 1$ 

$coev : 1 \longrightarrow X \otimes X^*$ 

s.t 

ok nice this way of picturing these morphisms gives us nice pictures. There is also another interesting morphism

if we have 2 vector spaces V, W then we know

$$V \otimes W \longrightarrow W \otimes V$$
$$v \otimes w \longmapsto w \otimes v$$

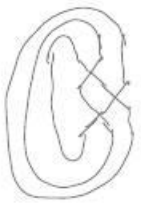
⇒ So there exist a swapping isomorphism 

$C_{V,W}$

s.t $C_{V,W} C_{W,V} = id$ these are called symmetric braiding which is also in Rep G for example.

Now, it "seems" that if we combine these pictures together we will draw knots

ex.



But the only problem we are facing here is this morphism

 it cannot distinguish between actual braidings of knots

 for example ?!

⇒ construct another category $\mathcal{Z}(\mathcal{C})$ which a morphism that behaves in this way!

$C \quad . \; Y \otimes X \longrightarrow X \otimes Y$

$C_{Y, X}$: ... → ... ⊗ ...





$C_{Y, X_1 \otimes X_2} = i_{12} \circ C_{Y, X_2} \circ C_{Y, X_1} \otimes id$

Similarly the other way around $C_{Y \otimes Y_2, X} = (\varsigma_{Y_1, X} \otimes id) \circ (id \otimes C_{Y_2, X})$

Rmk: this is called the hexagon diagram (due to associators which are not presented here)!

it turn out actully that we don't need to impose both relation actually just one of them, the other will be automatically satisfied by fixing $C_{Y, X}$ to be a module morphism.

Def: The Drinfeld center $Z(C)$ of a strict monoidal category C is the category of pairs $(Y, C_{Y,-})$ where

$$C_{Y,-} : Y \otimes - \longrightarrow - \otimes Y$$

is a functorial isomorphism $\forall X$ s.t the above relation

$$C_{Y, X_1 \otimes X_2} = (id \otimes C_{Y, X_2}) \circ (C_{Y, X_1} \otimes id)$$

$Z(C)$ is monoidal

$$(Y, C_Y) \otimes (Z, C_Z) = (Y \otimes Z, C_{Y \otimes Z})$$

Prop: $Z(C)$ is a braided monoidal category with a braiding $C_{Y,W}$

Thus, given $Rep(H)$ we constructed a category $Z(Rep(H))$ which can produce these invariants of knots!

Now 2 question!

1) Now that we have imposed this extra relation Does this induce some str on H itself, like it did with monoidality and rigidity?

2) is $Z(Rep(H))$ also an A-mod ?! Are these modules over some other algebra $D(H)$? How is $D(H)$ related to H?

Answer 1) let $C = H\text{-mod}$ be braided rigid monoidal
$F : C \longrightarrow Vec$ be it's exact faithful monoidal functor
(we will not require it to preserve the braiding we don't care)

then for $H = End\, F$.

we get a map in $\mathrm{End}(F\otimes F) \simeq H\otimes H$

$$F(X)\otimes F(Y) \xrightarrow{J} F(X\otimes Y) \xrightarrow{F(c)} F(Y\otimes X) \xrightarrow{J^{-1}} F(Y)\otimes F(X)$$

$$P\circ R$$

this is not a trivial swapping! R is invertible since c is and that it being a module morphism forces the following relation

$$R\,\Delta(a) = \Delta^{op}(a)\,R$$

the two hex axioms give us

$$(\Delta\otimes id)R = R^{13}R^{23}, \quad (id\otimes\Delta)R = R^{13}R^{12}$$

$R = \sum a_i\otimes b_i \qquad R_{12} = R\otimes 1 \qquad R_{23} = 1\otimes R$

$R_{13} = (P\otimes id)(1\otimes R)(P\otimes id)$

Def: An element $R\in H\otimes H$ satisfying these properties is called a quasi-tri str on H. R universal R matrix of H.

Prop: $\mathrm{Rep}H$ is braided iff H is quasi triangular.

Rmks: • the morphism $u \mapsto u$, $v \mapsto ?$ $\Leftrightarrow$ $h = S^2(h)$ for H-modules

$S^2(h) = h$ iff H is commutative or cocommutative.

- Some categories like spherical categories in QC are not tensor
- $Z(C)$ is a filter!
